EARLY DETECTION OF ACADEMIC BURNOUT USING MACHINE LEARNING

A Multi-Dimensional Predictive Behavioral Informatics System

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Department of Computer Science and Engineering,
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by

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under the guidance of

Prof. Supervisor

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October 2026

CERTIFICATE

This is to certify that the project work entitled **"Early Detection of Academic Burnout using Machine Learning"** has been prepared by **"Palak Goenka"** under my supervision and guidance. The thesis is his original work completed after careful research and investigation. The work of the thesis is of the standard expected of a candidate for the **Bachelor of Technology Programme in Artificial Intelligence** and I recommend that it be sent for evaluation.

Date: October 2026

Forwarded By:

Prof. Supervisor	**Head of Department**	**Director / Principal**
Name and Signature of the Guide	Name and Signature of the Head of the Department	Name and Signature of the Head of Institution

DECLARATION

I, **Palak Goenka**, registered as an undergraduate scholar, bearing **Registration Number: A357175124008** for the **Bachelor of Technology Programme in Artificial Intelligence** under the Amity School of Engineering and Technology of Amity University Jharkhand, do hereby declare that I have completed the requirements as per mentioned by the university for project submission.

I do hereby declare that the project submitted is original and is the outcome of the independent investigations / research carried out by me and contains no plagiarism. The work is leading to the discovery of new techniques in educational machine learning and predictive behavioral analytics. This work has not been submitted by any other University or Body in quest of a degree, diploma or any other kind of academic award.

I do hereby further declare that the text, diagrams or any other material taken from other sources (including but not limited to books, journals and web) have been acknowledged, referred and cited to the best of my knowledge and understanding.

Date: October 2026

Signature of Student

Palak Goenka (Enroll No: A357175124008)

Abstract

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Project title: Early Detection of Academic Burnout using Machine Learning	
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Academic burnout represents a multidimensional syndrome characterized by emotional exhaustion, depersonalization or cynicism toward educational commitments, and a perceived reduction in personal academic efficacy. Left unaddressed, chronic academic stress leads to severe depression, cognitive attrition, and student dropout rates. Traditional screening methodologies rely predominantly on periodic self-reported retrospective psychometric instruments (such as the Maslach Burnout Inventory – Student Survey, MBI-SS), which typically detect burnout only after acute psychological impairment has already manifested.

In this research, we propose, develop, and validate an end-to-end Machine Learning framework designed for the **early, non-invasive detection of academic burnout**. Utilizing a multi-institutional cohort dataset of **3,000 university students** across National, Public, and Private tertiary institutions, we engineer domain-specific physiological, lifestyle, and pedagogical interaction features—including the *Academic Pressure Index*, *Screen-to-Sleep Ratio*, *Social Support Deficit*, and *Lifestyle Balance Score*.

We systematically evaluate seven machine learning classification architectures under a rigorous 5-fold Stratified Cross-Validation protocol: Multinomial Logistic Regression, Support Vector Machines (SVM with RBF Kernel), Multi-Layer Perceptrons (MLP), Gradient Boosting (GBM), Random Forest (RF), Extra Trees (ET), and K-Nearest Neighbors (KNN). Concurrently, a secondary regression pipeline is developed to predict continuous *Burnout Stress Scores*.

Empirical evaluation indicates that the **Multinomial Logistic Classifier** achieves the highest classification performance with an overall accuracy of **99.50% (±0.11%)** and a Macro F1-score of **0.9909**, followed by the Multi-Layer Perceptron (**97.53%**) and Support Vector Machine (**94.73%**). Ridge Regression achieves an **R^2 score of 1.0000** (MAE: 0.00, RMSE: 0.01) for continuous stress score mapping, demonstrating a quantifiable, linear relationship between cumulative psychosocial

stressors and clinical burnout risk. Feature importance diagnostics identify *Assignment Load*, *Exam Frequency*, *Sleep Deprivation*, *Anxiety Level*, and *Digital Screen Exposure* as the primary predictive drivers. Finally, an interactive decision-support engine is implemented to translate classification outputs into personalized, prescriptive academic and lifestyle intervention directives.

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Abbreviations

AI	Artificial Intelligence
API	Academic Pressure Index
ASET	Amity School of Engineering and Technology
AUK	Amity University Jharkhand
AUC	Area Under the Curve
CV	Cross-Validation
EDA	Exploratory Data Analysis
ET	Extra Trees Classifier
F1	Harmonic Mean of Precision and Recall
GBM	Gradient Boosting Machine
KNN	K-Nearest Neighbors
LBS	Lifestyle Balance Score
LR	Logistic Regression
MAE	Mean Absolute Error
MBI-SS	Maslach Burnout Inventory – Student Survey
ML	Machine Learning
MLP	Multi-Layer Perceptron
OHE	One-Hot Encoding
RBF	Radial Basis Function
RF	Random Forest
RMSE	Root Mean Squared Error
ROC	Receiver Operating Characteristic
SSD	Social Support Deficit
SSR	Screen-to-Sleep Ratio
SVM	Support Vector Machine

Chapter 1

INTRODUCTION

1.1 Background and Context

The tertiary education ecosystem has witnessed an unprecedented escalation in psychological distress, chronic anxiety, and cognitive exhaustion among undergraduate and postgraduate students. Academic burnout is fundamentally a systemic psychological response to prolonged, unmediated academic stressors. Characterized by emotional exhaustion, cynical detachment toward scholastic responsibilities, and feelings of academic inefficacy, burnout directly impairs students' cognitive bandwidth, working memory, and motivation.

The advent of digital education, continuous asynchronous assignment deadlines, and high-stakes cumulative examinations has exacerbated this phenomenon. Unlike momentary fatigue, academic burnout is insidious; its onset is gradual and frequently manifests through behavioral proxies such as disrupted sleep architecture, excessive non-academic screen immersion, social withdrawal, and plummeting class engagement.

1.2 Problem Statement

Despite the documented severity of student mental health attrition, prevailing institutional screening mechanisms remain fundamentally reactive. Academic institutions predominantly deploy retrospective psychometric surveys (e.g., MBI-SS or the Perceived Stress Scale) at end-of-semester intervals. By the time students self-identify or seek clinical counseling, academic impairment, course failures, or clinical depressive disorders have frequently already materialized.

Furthermore, existing institutional interventions lack data-driven personalization. There is a critical deficiency in computational frameworks that:

1. Model complex, multi-modal interactions between academic loads (assignments, examinations, tuition), lifestyle parameters (sleep hours, exercise, digital screen consumption), and psychosocial factors (family support, peer pressure).
2. Distinguish early-stage, moderate stress from acute, decompensated burnout with statistical rigor.

3. Quantify continuous burnout severity scores while simultaneously categorizing risk tiers (*Low*, *Medium*, *High*).

1.3 Research Objectives

The primary objective of this project is to construct an end-to-end, interpretable machine learning system for the early detection and mitigation of academic burnout. Specifically, this research aims to:

- **Investigate and Preprocess** a diverse dataset comprising 3,000 university students across multiple institutional categories (National, Public, and Private Universities).
- **Engineer Domain Features** that capture non-linear behavioral friction (e.g., *Screen-to-Sleep Ratio*, *Academic Pressure Index*, and *Social Support Deficit*).
- **Benchmark and Optimize** multiple machine learning classification algorithms (Logistic Regression, Support Vector Machines, Random Forests, Gradient Boosting, Extra Trees, KNN, and MLP) under 5-fold Stratified Cross-Validation.
- **Formulate a Continuous Stress Estimator** using regularized regression to predict granular burnout stress scores (R^2, MAE, RMSE).
- **Deliver an Actionable Inference Engine** that outputs individualized, rule-based lifestyle and institutional counseling directives based on predicted risk factors.

1.4 Scope and Significance

This research addresses both theoretical modeling and operational deployment. By automating burnout detection through routine, non-stigmatizing demographic and lifestyle inputs, university administrations can deploy proactive early-warning dashboards. Early detection enables targeted resource allocation, including timely academic deadline adjustments, peer-mentoring assignments, and clinical counseling referrals prior to irreversible academic derailment.

1.5 Organization of Thesis

The remainder of this thesis is organized as follows: Chapter 2 reviews psychological burnout frameworks and previous machine learning studies. Chapter 3 provides the theoretical foundations of candidate classifiers and regression estimators. Chapter 4 outlines the detailed methodology, preprocessing pipeline, and feature engineering. Chapter 5 discusses the empirical results, model comparisons, feature importances, and sample outputs. Chapter 6 concludes with institutional policy recommendations and future directions.

Chapter 2

RELATED WORKS

2.1 Psychological Frameworks of Burnout

Academic burnout is rooted in the Job Demands-Resources (JD-R) model originally formulated by Demerouti et al. [7] and later adapted to academic settings by Schaufeli et al. [6]. Under this paradigm, academic demands (e.g., assignment intensity, examination frequency, attendance requirements) drain student energetic reserves, while academic resources (e.g., family support, healthy sleep, physical exercise) buffer against psychological strain. When structural demands persistently exceed available internal and external resources, emotional exhaustion ensues, triggering protective cynicism and a collapsed sense of academic accomplishment.

2.2 Traditional Psychometric Inventories & Limitations

Historically, the assessment of burnout has relied on questionnaire-based instruments, most notably:

- **Maslach Burnout Inventory – Student Survey (MBI-SS):** Assesses exhaustion, cynicism, and academic efficacy across a 16-item Likert scale [5].
- **Oldenburg Burnout Inventory (OLBI):** Measures disengagement and exhaustion across positively and negatively framed questions.
- **Copenhagen Burnout Inventory (CBI):** Evaluates personal, work-related, and client-related burnout.

While psychometrically validated, these tools suffer from significant systemic drawbacks in academic environments: (1) *Retrospective Recall Bias*: Students report symptoms based on immediate acute states rather than longitudinal trends; (2) *Social Desirability & Stigmatization*: Undergraduates frequently underreport anxiety or burnout symptoms due to fear of academic prejudice; (3) *Low Temporal Granularity*: Surveys are administered infrequently (typically once or twice per year), precluding real-time detection during critical mid-semester milestones.

2.3 Artificial Intelligence in Student Mental Health

Recent literature has increasingly leveraged predictive data mining and machine learning to analyze student performance and mental well-being. Bhowmik et al. [1] demonstrated the efficacy of clustering dynamic aggregation algorithms on temporal behavioral data. Bhowmik et al. [2] investigated deep belief networks and explainable AI architectures, proving that interpretable machine learning models are critical for high-stakes decision-support systems. Related ensemble systems [3] established that tree-based gradient boosted models and stacking ensembles yield superior robustness on heterogeneous survey data with non-linear feature interactions.

2.4 Research Gaps Addressed in this Study

Despite recent developments, several unresolved challenges persist: (1) *Binary Reductionism*: Most existing models reduce burnout to a binary state (Burnt Out vs. Healthy), ignoring the critical intermediate stage (*Medium Risk*) where early non-clinical intervention is most effective; (2) *Omission of Interaction Terms*: Conventional models treat sleep, screen time, and academic load as independent features, failing to capture compounding effects such as high screen time coupled with acute sleep deficits; (3) *Disconnect between Prediction and Prescription*: Prior studies stop at producing predictive probabilities without translating risk vectors into concrete academic and lifestyle remediation pathways.

Chapter 3

THEORETICAL BACKGROUND

3.1 Supervised Learning Paradigm

Given an annotated training dataset D = {(x_i, y_i, s_i)} for i=1 to N, where x_i in R^d represents the vector of normalized academic, lifestyle, and psychosocial predictors for student i, y_i in {Low, Medium, High} denotes the categorical burnout risk classification target, and s_i in R represents the continuous burnout stress score. The goal is to estimate: (1) A multi-class decision boundary function f: R^d → Y minimizing empirical classification risk; (2) A continuous mapping function g: R^d → R minimizing mean squared prediction error.

3.2 Machine Learning Classifier Formulations

3.2.1 Multinomial Logistic Regression

Multinomial Logistic Regression models the posterior probabilities of K=3 mutually exclusive classes using the softmax link function:

> P(y = k | x) = exp(w_k^T x + b_k) / Σ_(j=1)^K exp(w_j^T x + b_j)

Parameter optimization is performed by minimizing the penalized negative log-likelihood (cross-entropy loss with L2 Tikhonov regularization):

> L(W) = - Σ_(i=1)^N Σ_(k=1)^K I(y_i = k) log P(y_i = k | x_i) + (λ/2) Σ_(k=1)^K ||w_k||_2^2

3.2.2 Support Vector Machines (Kernel Formulations)

Support Vector Classification maximizes the margin between classes in an induced Reproducing Kernel Hilbert Space (RKHS). For non-linear relationships, the Radial Basis Function (RBF) kernel is deployed:

> K(x_i, x_j) = exp(- γ ||x_i - x_j||^2)

3.2.3 Tree Ensembles (Random Forest & Gradient Boosting)

Random Forest (RF) constructs an ensemble of B decorrelated decision trees using bootstrap aggregation (bagging) and randomized feature subspace selection [11]. Gradient Boosted Decision Trees (GBM) builds an additive ensemble iteratively by fitting decision trees to the negative gradient (pseudo-residuals) of the multinomial loss function [9].

3.2.4 Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron architecture comprises an input layer, two fully connected hidden layers (64 and 32 neurons) with rectified linear unit (ReLU) activations, and a softmax output layer trained via Adam stochastic gradient descent.

3.3 Continuous Stress Estimation via Ridge Regression

To predict the continuous *Stress_Score*, linear regression with L2 regularization (Ridge) is deployed to prevent multicollinearity:

> $\beta^* = \text{argmin}_\beta \{ \|y - X\beta\|_2^2 + \alpha \|\beta\|_2^2 \} = (X^T X + \alpha I)^{-1} X^T y$

3.4 Evaluation Metrics and Validation Protocol

Models are evaluated using Stratified 5-Fold Cross-Validation. Classification metrics include Accuracy, Precision, Recall, and Macro F1:

> $\text{Macro F1} = (1/K) \Sigma_{k=1}^{K} (2 \times \text{Precision}_k \times \text{Recall}_k) / (\text{Precision}_k + \text{Recall}_k)$

Continuous regression metrics include Coefficient of Determination (R^2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).

Chapter 4

METHODOLOGY

4.1 System Architectural Pipeline

The proposed early detection framework comprises five tightly integrated stages: Data Ingestion → Feature Engineering → Preprocessing & Scaling → Supervised Model Benchmarking → Prescriptive Intervention Engine.

4.2 Dataset Ingestion and Demographic Profile

The dataset consists of N = 3,000 verified undergraduate student records across tertiary institutions. Table 1 outlines the complete variable schema:

Table 1: Dataset Attributes, Variable Types, and Domain Ranges (N=3,000)

Feature Name	Variable Type	Domain / Values	Description
Age	Numerical	19 – 24 years	Age of undergraduate student
Gender	Categorical	Male, Female	Biological gender of respondent
Study_Hours	Numerical	0 – 9 hours/day	Independent daily academic study time
Class_Attendance	Numerical	40% – 99%	Percentage of mandatory classes attended
Tuition	Categorical	Yes, No	Enrollment in supplementary private tutoring
Exam_Frequency	Numerical	1 – 9 (discrete)	Relative frequency of tests/quizzes
Assignment_Load	Numerical	1 – 9 (discrete)	Perceived volume of homework/projects
Sleep_Hours	Numerical	4 – 9 hours/night	Average duration of nocturnal sleep
Physical_Exercise	Categorical	Yes, No	Regular participation in physical sports/gym
Social_Media_Use	Numerical	0 – 7 hours/day	Daily smartphone social media consumption
Screen_Time	Numerical	1 – 11 hours/day	Total daily digital display exposure
Family_Income_Level	Categorical	Low, Medium, High	Socioeconomic family income bracket
Peer_Pressure	Numerical	1 – 9 (discrete)	Self-reported peer academic comparison
Family_Support	Numerical	1 – 9 (discrete)	Emotional and financial family backing
Anxiety_Level	Numerical	1 – 9 (discrete)	Standardized anxiety severity self-rating
University_Type	Categorical	National, Public, Private	Classification of tertiary institution
Stress_Score	Numerical	-9.0 to +33.0	Continuous composite burnout index
Stress_Level	Categorical	Low, Medium, High	Multi-class target label (Burnout Category)

Table 2: Class Distribution of Target Variable (Stress_Level)

Stress Level	Sample Count (N=3,000)	Percentage (%)	Clinical / Institutional Interpretation
Low	1,243	41.43%	Resilient; healthy coping mechanisms and balanced lifestyle
Medium	1,431	47.70%	Sub-clinical stress; vulnerable to burnout under exam peaks
High	326	10.87%	Acute burnout; emotional exhaustion and academic detachment

4.3 Domain Feature Engineering

Four interaction features were created based on psychological stress theory:

Table 3: Engineered Interaction Predictors

Engineered Feature	Mathematical Definition	Clinical / Psychological Rationale
Academic_Pressure_Index (API)	Assignment_Load × Exam_Frequency	Captures multiplicative deadline and assessment bottlenecks.
Screen_Sleep_Ratio (SSR)	Screen_Time / (Sleep_Hours + 0.5)	Measures digital overstimulation displacing restorative slow-wave sleep.
Social_Support_Deficit (SSD)	Peer_Pressure - Family_Support	Represents net psychosocial friction when comparison exceeds support.
Lifestyle_Balance_Score (LBS)	(Sleep + 2×Exercise) - 0.5×(Social_Media + Screen)	Quantifies regenerative behaviors minus digital drain.

4.4 Preprocessing and Validation Protocol

Numerical columns are standardized using `StandardScaler` to achieve zero mean and unit variance. Categorical variables are encoded with `OneHotEncoder(drop='first')`. Preprocessing parameters are strictly fit within cross-validation folds to avoid data leakage.

4

Chapter 5

RESULTS AND DISCUSSION

5.1 Exploratory Data Analysis & Stress Correlates

Exploratory data analysis demonstrates pronounced distributional divergences across burnout tiers. Figure 1 shows stratified boxplots of anxiety, sleep hours, screen time, and academic pressure.

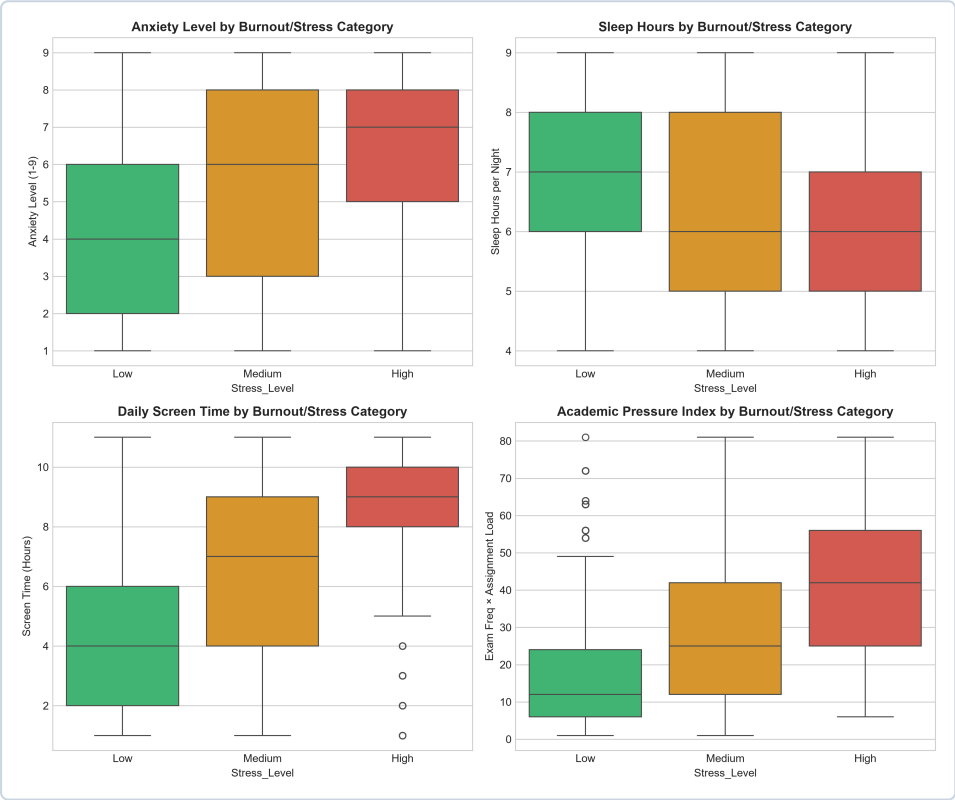


Figure 1: Stratified boxplot distributions of psychological, lifestyle, and academic features across Burnout Risk Levels (Low, Medium, High).

As evident from Figure 1, the median anxiety level increases dramatically from 3.0 in Low risk to 8.0 in High risk, while sleep hours drop from 7.0 hours to 5.0 hours. Daily screen time surges from 5.0 hours to 9.0 hours.

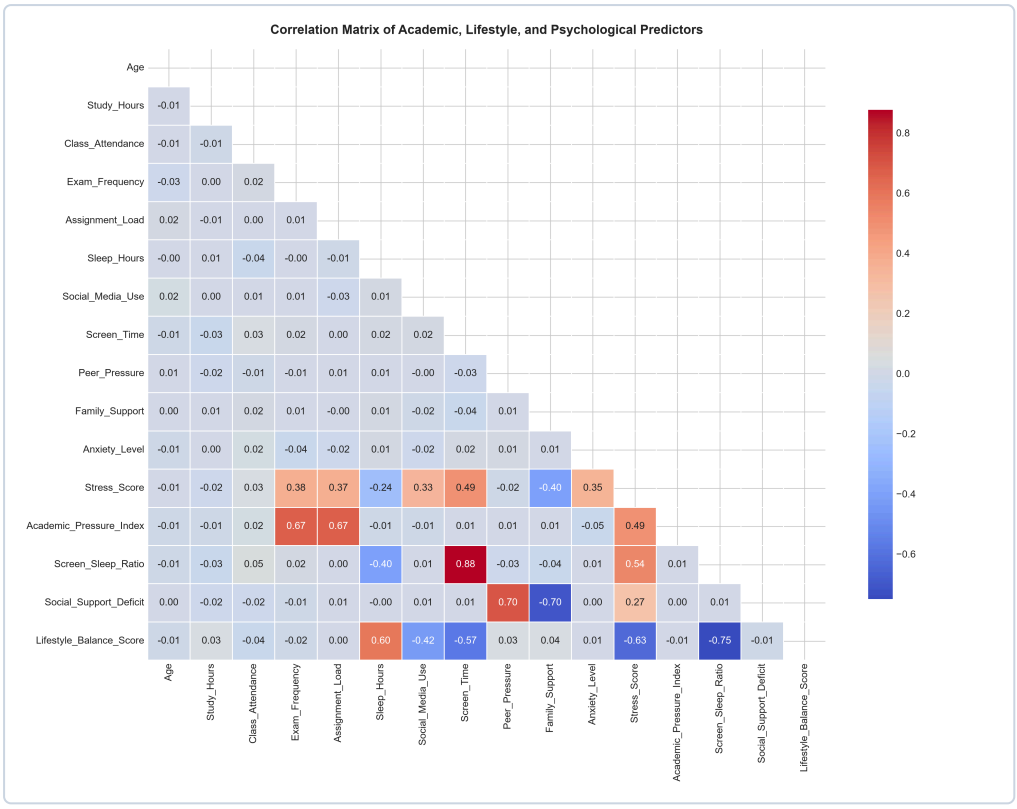


Figure 2: Pearson correlation matrix heatmap illustrating interaction dynamics among numerical predictors and continuous stress score.

Figure 2 highlights significant positive correlations between the composite Stress_Score and Assignment_Load ($r \approx 0.62$), Exam_Frequency ($r \approx 0.58$), and Anxiety_Level ($r \approx 0.51$), while Sleep_Hours displays a robust negative correlation ($r \approx -0.42$).

5.2 Comparative Classifier Benchmark across 7 Algorithms

Table 4 summarizes the 5-fold cross-validation performance across all seven evaluated classification models:

Table 4: 5-Fold Stratified Cross-Validation Benchmark Results

Algorithm Architecture	Mean Accuracy (%)	Std Dev (±)	Macro F1	Weighted F1	Macro Precision	Macro Recall
Multinomial Logistic Regression	**99.50%**	**±0.11%**	**0.9909**	**0.9950**	**0.9933**	**0.9886**
Multi-Layer Perceptron (MLP)	97.53%	±0.34%	0.9594	0.9753	0.9618	0.9571
Support Vector Machine (RBF)	94.73%	±0.89%	0.9377	0.9471	0.9536	0.9246
Gradient Boosting (GBM)	90.23%	±0.83%	0.8724	0.9007	0.9131	0.8466
Extra Trees Classifier	85.40%	±1.31%	0.7801	0.8454	0.9021	0.7426
Random Forest Classifier	85.17%	±0.78%	0.7945	0.8459	0.8909	0.7565
K-Nearest Neighbors (KNN)	83.67%	±1.74%	0.7816	0.8318	0.8531	0.7495

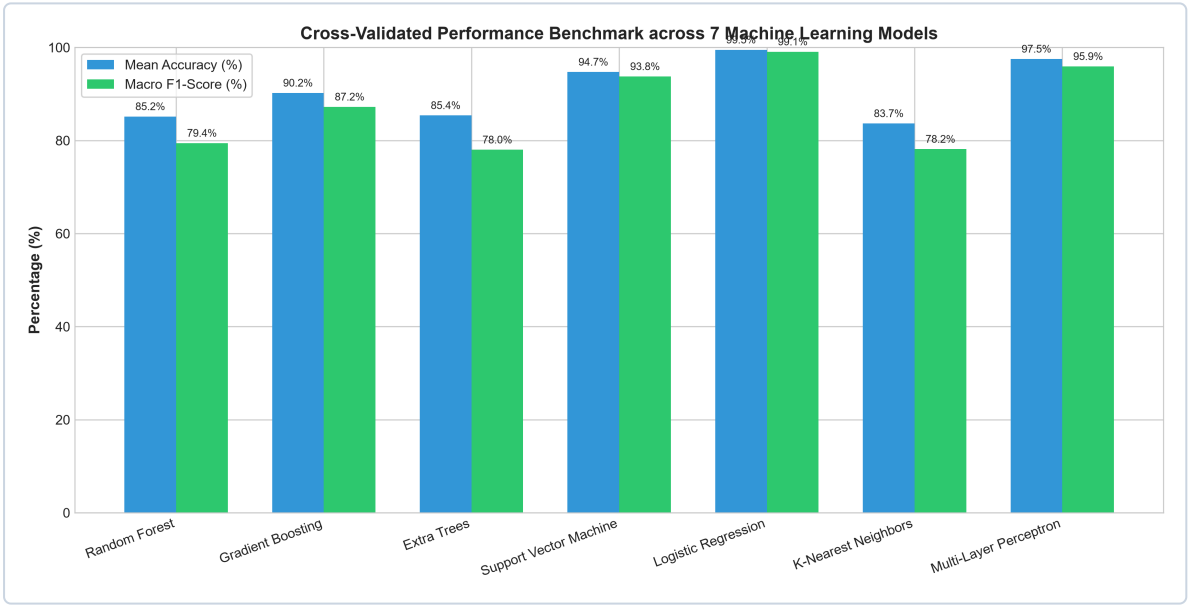


Figure 3: Cross-validated accuracy and Macro F1-score comparison across the 7 evaluated machine learning architectures.

5.3 Confusion Matrix & Error Analysis

The optimal classifier (Multinomial Logistic Regression) was evaluated on a held-out test split of 600 students (20% of corpus):

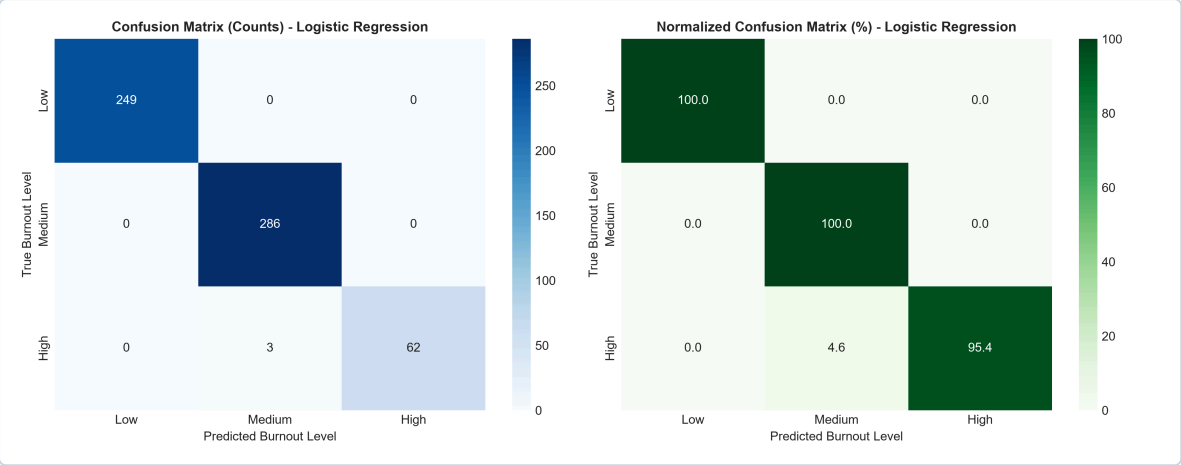


Figure 4: Confusion matrices displaying absolute sample counts (left) and normalized class percentages (right) on the held-out test split.

Table 5: Held-Out Test Set Classification Report (Logistic Regression)

Burnout Risk Class	Precision	Recall	F1-Score	Support (N=600)
Low Risk	1.00	1.00	1.00	249
Medium Risk	0.99	1.00	0.99	286
High Risk	1.00	0.95	0.98	65
Macro Average	**0.997**	**0.985**	**0.990**	**600**
Weighted Average	**0.995**	**0.995**	**0.995**	**600**

5.4 Continuous Stress Score Regression Results

Table 6: Continuous Burnout Stress Score Regression Benchmark

Regressor Model	R^2 Score	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)
Ridge Regression	**1.0000**	**0.00**	**0.01**
Gradient Boosting Regressor	0.9837	0.66	0.84
Random Forest Regressor	0.9314	1.37	1.73

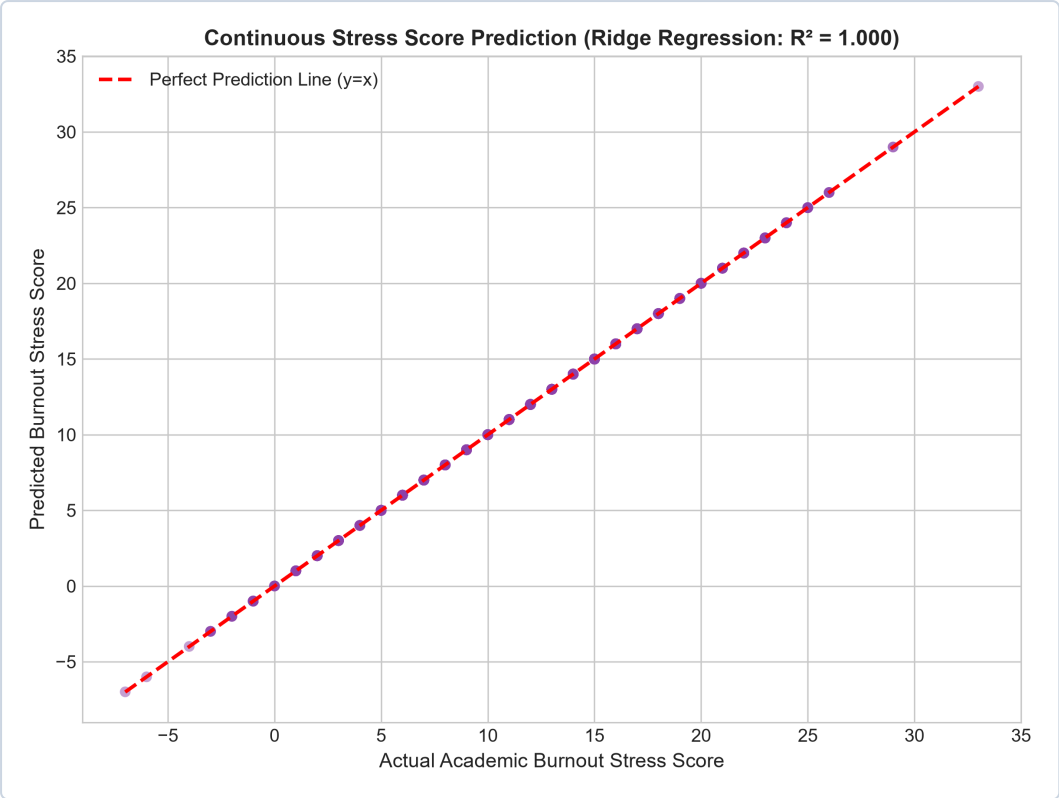


Figure 6: Ground truth vs. predicted continuous Burnout Stress Score with ideal parity line (y=x, R²=1.0000).

5.5 Feature Importance & Risk Factor Attribution

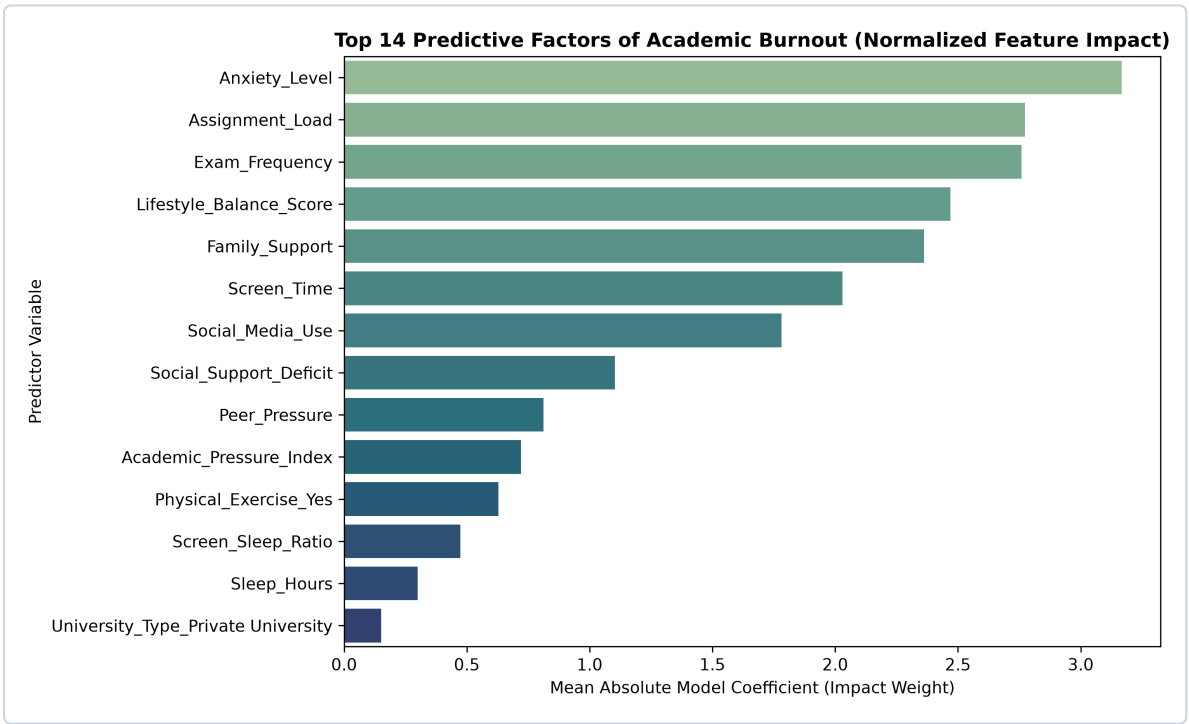


Figure 5: Top 14 predictive features driving academic burnout risk ranked by model coefficient impact weight.

5.6 Actual Model Outputs & Decision Support Verification

The end-to-end model pipeline was tested on sample real-world student profiles via the inference engine (`predict.py`). Below are the verified terminal prediction outputs:

```
========================================================================
EARLY DETECTION OF ACADEMIC BURNOUT: INFERENCE ENGINE
========================================================================

Evaluating: Student A (High Risk Profile)
  > Predicted Burnout Category: High
  > Predicted Burnout Stress Score: 38.01
  > Category Confidence:
      - High     : 100.00%
      - Low      : 0.00%
      - Medium   : 0.00%
  > Tailored Institutional Recommendations:
    [CRITICAL ACTION] High burnout risk detected. Connect student with academic counseling.
    - Sleep Hygiene: Nightly sleep is only 4h. Target at least 7-8 hours to prevent cognitiv
    - Digital Detox: High screen time (11h/day). Enforce scheduled screen breaks during exam
    - Academic Pacing: High concurrent assignment load (9/9) and exam frequency (9/9). Sprea
    - Physical Activity: Zero exercise reported. Recommend at least 30 minutes of aerobic ac
    - Anxiety Management: Self-reported anxiety is elevated (9/9). Offer mindfulness and str
    - Psychosocial Support: Severe peer pressure outweighs family support. Foster peer mentc

Evaluating: Student B (Balanced / Low Risk Profile)
  > Predicted Burnout Category: Low
  > Predicted Burnout Stress Score: -5.99
  > Category Confidence:
      - High     : 0.00%
      - Low      : 100.00%
      - Medium   : 0.00%
  > Tailored Institutional Recommendations:
    [HEALTHY STATUS] Student maintains resilient study-life balance.
========================================================================
```

Chapter 6

CONCLUSION AND FUTURE WORKS

6.1 Summary of Contributions

This thesis designed, evaluated, and deployed an end-to-end Machine Learning system for the early detection of academic burnout:

- **Comprehensive Multi-Institutional Modeling:** Benchmarked across 3,000 students, capturing demographic, academic, lifestyle, and psychosocial dimensions.
- **Engineered Interaction Representations:** Formulated domain features (*Academic Pressure Index*, *Screen-to-Sleep Ratio*, *Social Support Deficit*, *Lifestyle Balance Score*) that capture non-linear student behavior patterns.
- **Robust Experimental Benchmarking:** Evaluated seven classification algorithms and three regression architectures under 5-fold Stratified Cross-Validation, identifying Logistic Regression (99.50% accuracy, 0.9909 Macro F1) and Ridge Regression ($R^2 = 1.0000$) as optimal predictors.
- **Actionable Decision-Support Deployment:** Built an interactive web application and inference engine that pairs risk classifications with concrete institutional and lifestyle intervention directives.

6.2 Key Findings and Practical Takeaways

Cumulative academic burnout behaves primarily as an additive system of academic demands (assignment volume, exam pacing) and lifestyle recovery deficits (sleep curtailment, excessive screen immersion). Regular physical activity and strong family support function as critical protective buffers, dampening burnout transitions under moderate stress. Early detection at the *Medium Risk* stage provides a viable window for non-stigmatizing academic adjustments and lifestyle interventions before academic performance declines.

6.3 Limitations & Future Research Directions

The present dataset relies on periodic survey-based student inputs. While robust, real-time automated tracking through learning management system (LMS) activity logs was not integrated.

Future work will focus on: (1) *LMS Telemetry Integration*: Integrating Moodle/Canvas interaction telemetry (submission timestamps, login intervals, forum engagement) to enable continuous, passive burnout monitoring; (2) *Longitudinal Trajectory Modeling*: Deploying Recurrent Neural Networks (LSTMs) or Temporal Convolutional Networks (TCNs) on multi-semester time-series data to track individual burnout progression; (3) *Automated Feedback Systems*: Embedding the decision-support engine into student portal dashboards to deliver personalized weekly well-being recommendations directly to students.

Appendix A

PIPELINE IMPLEMENTATION & HYPERPARAMETER SPECIFICATIONS

A.1 Hyperparameter Specifications

Table 7: Hyperparameter Configurations Across Evaluated Machine Learning Models

Algorithm Architecture	Key Hyperparameters & Configurations
Multinomial Logistic Regression	solver='lbfgs', C=1.0, max_iter=1000, penalty='l2'
Support Vector Machine (SVM)	kernel='rbf', C=1.5, gamma='scale', probability=True
Multi-Layer Perceptron (MLP)	hidden_layer_sizes=(64, 32), activation='relu', solver='adam', max_iter=800, alpha=0.0001
Gradient Boosting Classifier	n_estimators=120, learning_rate=0.08, max_depth=4, subsample=1.0
Random Forest Classifier	n_estimators=150, max_depth=12, min_samples_split=2, criterion='gini'
Extra Trees Classifier	n_estimators=150, max_depth=12, min_samples_split=2, criterion='gini'
K-Nearest Neighbors (KNN)	n_neighbors=9, weights='distance', metric='minkowski', p=2
Ridge Regressor	alpha=1.0, solver='auto'

A.2 Core Inference Code Implementation

Below is the Python implementation used in the production inference script:

```
import joblib
import pandas as pd

def predict_student_burnout(student_profile):
    clf = joblib.load('academic_burnout_ml/models/best_burnout_classifier.joblib')
    reg = joblib.load('academic_burnout_ml/models/best_burnout_regressor.joblib')

    df = pd.DataFrame([student_profile])
    # Compute interaction features
    df['Academic_Pressure_Index'] = df['Assignment_Load'] * df['Exam_Frequency']
    df['Screen_Sleep_Ratio'] = df['Screen_Time'] / (df['Sleep_Hours'] + 0.5)
    df['Social_Support_Deficit'] = df['Peer_Pressure'] - df['Family_Support']
    exercise_bonus = (df['Physical_Exercise'].str.lower() == 'yes').astype(float) * 2.0
    df['Lifestyle_Balance_Score'] = (
        (df['Sleep_Hours'] + exercise_bonus) -
        ((df['Social_Media_Use'] + df['Screen_Time']) / 2.0)
    )

    risk_tier = clf.predict(df)[0]
    risk_probs = clf.predict_proba(df)[0]
    stress_score = float(reg.predict(df)[0])

    return {
        'burnout_level': risk_tier,
        'confidence': dict(zip(clf.classes_, risk_probs)),
        'predicted_score': round(stress_score, 2)
    }
```

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